

ERGT: Attention-Free Native Geodesic Transport for Controlled Long-Horizon Relational Reasoning

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Abstract

We introduce Executable Relational Geometry and Transport (ERGT), an attention-free architecture in which relational computation is organized by input-conditioned geometry rather than self-attention. ERGT learns a mapping from raw tokens to a persistent information field Ψ , whose pairwise observables induce multiple directed world geometries. These geometries then become the computational substrate for reasoning: GeoTransport carries typed state along admissible paths, and a geodesic boundary solver accepts the least-cost candidate consistent with propagation, memory, payload, and terminal constraints. This native answer path uses neither a Transformer nor a supplied graph, target compiler, or generic executor.

Across four independent runs, ERGT converged and satisfied every registered native stability contract. Beyond its training range, mean accuracy at 32 hops was 0.955, compared with 0.500 for a strengthened direct Transformer given the same raw input and a closely matched parameter count. Frozen same-checkpoint interventions caused sharp losses when geodesic action, propagation admissibility, typed transport, geometric memory, or terminal measurement was removed. Randomizing or shuffling the geometry, or restricting inference to one world, also failed to recover the intact result. The evidence supports a bounded architectural claim: a Ψ field formed from raw tokens can become an executable geometry and provide a measurable advantage in controlled long-horizon relational reasoning.

Keywords: relational reasoning; representation learning; information geometry; geodesic transport; causal intervention; multiworld geometry; attention-free neural architecture; long-horizon generalization

1 Introduction

Modern sequence models have shown how effective content-dependent token interaction can be for representation learning. In the Transformer, that interaction is implemented by self-attention [1].

Code availability. The complete source code and executable reproduction package accompanying this article, including the Colab notebook, are publicly available at github.com/jalaljafari2009/ERGT-paper.

However, long-horizon relational reasoning imposes additional structural requirements. Entities must remain distinguishable, directed and typed relations must compose in the correct order, intermediate state must survive repeated propagation steps, and the model should decline to answer when no valid support path exists. These requirements can be learned implicitly, but a generic sequence backbone does not make admissibility, conservation, or rejection part of the computation itself.

Executable Relational Geometry and Transport (ERGT) is built around that distinction. Its central premise is that the arrangement of information in an input can generate the geometry on which reasoning takes place. Rather than starting from attention scores or a graph supplied with the task, the model forms a persistent information field Ψ from the raw tokens. This field keeps identity and relational evidence tied to the same source state, providing the substrate from which all later geometry is derived.

Pairwise measurements of Ψ produce several directed geometries, or *worlds*, each acting as a lens on a different relational tendency. The term is operational: a world contributes support, direction, and traversal cost; it does not denote a literal physical universe. GeoTransport propagates typed state along the resulting admissible paths while preserving entity, operator, payload, and memory information. A native geodesic solver then finds the least-cost valid route, and boundary measurement accepts a candidate only when the transported state satisfies the terminal constraints. The complete path is

$$\begin{aligned} \text{raw tokens} &\longrightarrow \Psi \longrightarrow \text{multiworld geometry} \\ &\longrightarrow \text{GeoTransport} \longrightarrow \text{geodesic solution} \longrightarrow \text{boundary decision.} \end{aligned}$$

No graph or geometry is supplied at the input boundary, and the tested native path does not delegate its decision to a generic compiler or graph executor. The relation support, path costs, transported state, and acceptance conditions must all arise from the field constructed from the same raw sequence.

The design is motivated by a specific long-horizon failure mode. When a solution depends on many typed relations, repeated global remixing may blur which state belongs to which entity and which path. ERGT instead treats composition as repeated local transport under shared admissibility and conservation rules. Its inductive bias is not simply a wider context window; it is the preservation of executable relational paths as reasoning depth increases.

The experimental evaluation follows the same logic. It first tests whether the full route from raw-token field formation to terminal measurement can converge and satisfy the registered native stability contracts. Frozen checkpoints are then evaluated on freshly generated, raw-disjoint relational chains well beyond the training horizon. Finally, frozen targeted interventions, geometry corruption, and single-world restrictions test whether the solved answers actually depend on the proposed mechanism.

The primary baseline is a strengthened direct Transformer trained on the same raw sequences and matched target families. Staged training and auxiliary supervision improve its opportunity to learn the task, while its inference path remains purely Transformer-based: it receives no ERGTgraph, geometry, transport state, compiler, executor, or identity side channel. This preserves a clear comparison between two complete inference strategies rather than lending the proposed solver to the baseline.

The contributions of this work are:

1. a persistent Ψ field, built from raw tokens, that supplies the internal identity and pairwise evidence from which relation geometry is induced rather than given;
2. an end-to-end, attention-free answer path combining directed multiworld lenses, finite-speed admissibility, typed GeoTransport, geometric memory, min-plus geodesic closure, terminal boundary measurement, and fail-closed output;
3. a controlled multi-seed protocol that separates learnability from long-horizon extrapolation while comparing ERGT with a strengthened direct Transformer at the same raw-input boundary; and
4. frozen causal interventions and a self-contained reproduction package that make the roles of geometry, transport, memory, boundary conditions, and multiworld composition independently auditable

and reproducible.

The claim remains deliberately bounded. ERGT is not offered as a universal replacement for Transformers, a general language model, or a cheaper sequence architecture. The study tests whether geometry induced from raw tokens can act as an executable and mechanistically testable bias for controlled long-horizon relational reasoning. Section 5 reports the corresponding outcomes, uncertainty estimates, and effect sizes.

2 Positioning and Related Work

2.1 Attention and attention-free sequence backbones

Removing attention is not, by itself, a new research direction. Self-attention computes content-conditioned interactions across sequence positions [1], whereas S4 realizes long-range propagation through a structured state-space system [2]. Mamba makes that state-space dynamics input-selective [3], and Hyena uses implicit long convolutions with data-controlled gates [4]. These models show that a sequence can be processed effectively without QKV attention.

Their main object is still the sequence operator: recurrence, state-space dynamics, or convolution determines how information is mixed over position. ERGT addresses a different object. It first constructs a token-indexed field and asks that field to support a directed, executable relation geometry. Only then can typed state propagate toward a boundary decision. The contribution is therefore not attention removal alone, nor a claim about language-modeling quality or linear-time efficiency, but a raw-input geometry whose constraints participate directly in inference.

2.2 Graph reasoning, path methods, and latent structure

Graph networks expose relational bias through message passing [5, 6], and geometric deep learning provides the broader language for computation on non-Euclidean structure [7]. Usually, however, the relevant nodes and edges arrive with the input. NBFNet is especially close in spirit because it represents a node pair through paths and learns a generalized Bellman–Ford computation [10]; its paths still belong to a graph that is already given.

Other work learns structure itself. Neural Relational Inference, for example, infers an interaction graph while predicting dynamics from observations [11]. The claim for ERGT is therefore more specific than “learning a graph.” Structure must arise from raw tokens and immediately become executable through directed metrics, path action, typed transport, admissibility, memory, and terminal measurement. The same distinction defines the role of Ψ : it is not an extra embedding layer but the audited source from which identity, pairwise evidence, and every world geometry must descend. A supplied edge, external entity identifier, or target graph would break that contract.

2.3 Neural algorithmic reasoning and size generalization

The CLRS benchmark made size generalization a central test of learned algorithm execution [8]. Hint-ReLIC likewise shows why intermediate computation matters, regularizing states to remain invariant under interventions that should preserve the next algorithmic step [9]. Both lines of work caution against treating final accuracy as sufficient evidence of a reusable computation.

ERGT adopts that evaluation principle without imitating a supplied trace or set of hints. It learns its geometric dynamics from the sequence and tests whether the same local transport and admissibility rules continue to compose at greater depth. After checkpoint selection, named geometric quantities are removed, shuffled, or restricted while the parameters remain frozen. The resulting change asks

whether a quantity was needed by the learned answer path, not merely whether supervising it helped during training.

2.4 Physics-informed inductive bias and information geometry

Physics-informed models illustrate the value of turning a desired law into an architectural restriction rather than hoping it appears as a correlation. Hamiltonian Neural Networks encode conservation in learned dynamics [14], while explicit Hamiltonian and Lagrangian constraints extend the same idea [15]. Geometric deep learning uses symmetry and geometry to restrict the hypothesis space [7]; information geometry supplies the language of metrics and dual structure on statistical manifolds [13]. ERGT adopts this as a geometric design analogy rather than a direct statistical-manifold construction.

ERGT borrows this design logic, not a literal physical interpretation. Its terms have concrete computational meanings: Ψ is persistent token information, worlds are directed relation lenses, action is traversal cost, the cone is a propagation constraint, and GeoTransport moves typed state. A geodesic is the least-cost admissible route, conservation protects payload identity, and boundary measurement determines acceptance. Classical shortest-path recursion and dynamic programming motivate the closure [16, 17].

Spectral graph quantities [18] and a Forman-style curvature proxy [19] are kept outside that answer path. They describe how the induced geometry changes with horizon and motivate future work on curvature-guided routing and regularization [12], as well as adaptive worlds; they are not used to attribute the current performance.

2.5 Position of the claimed contribution

Table 1 brings these boundaries together. The decisive difference is not any single familiar component, but the object the model must construct and execute.

No isolated ingredient carries the novelty claim: shortest paths, graph learning, conservation, attention-free propagation, and boundary conditions all have established precedents. What is tested here is their integration under one auditable contract. Raw tokens form a persistent Ψ field; the field induces several directed geometries; hard constraints govern typed transport; an admissible geodesic is composed; and only terminal measurement can accept an answer. This geometry-first contract, rather than general superiority on sequence tasks, defines the proposed advantage and the empirical burden of the paper.

3 Method: From Raw Tokens to Executable Geometry

3.1 Conceptual design

The geometry-first idea becomes useful only if it defines the entire route to an answer. That route has three responsibilities. It must infer relational structure from raw symbols without an external graph or identity map; compose local relations without losing the entity, operator, or payload in transit; and distinguish a supported terminal state from a candidate that is merely reachable.

These responsibilities lead to three connected operations. *Geometry formation* builds the persistent Ψ field, measures pairwise evidence, and induces several directed world metrics. *Geometric execution* converts those metrics into admissible actions and moves typed state through their product space. *Boundary adjudication* accepts the resulting path only if the correct register, conserved payload, and terminal conditions arrive together. Geometry is therefore part of the decision process, not an auxiliary visualization: the output of each operation is the state or constraint required by the next.

Table 1: Position of ERGTrelative to neighboring research families.

Family	Primary computational object	Boundary with ERGT
Transformer	Content-conditioned token mixing	Does not explicitly require admissible paths, typed state conservation, or fail-closed boundary evidence.
S4, Mamba, Hyena	State-space recurrence or long convolution over a sequence	Replace the sequence operator; they do not by themselves construct and execute a relation geometry from raw tokens.
GNNs and NBFNet	Message passing or path aggregation on a graph	Usually receive the graph; ERGTmust induce support and metric before transporting typed state.
Latent graph inference	Inferred interaction structure and predictive dynamics	ERGTradditionally turns induced structure into multiworld action, admissibility, transport, memory, and a boundary decision.
CLRS and Hint-ReLIC	Learned algorithm execution and invariant intermediate computation	ERGTrlearns and intervenes on a native geometric dynamics rather than a supplied algorithm trace.
Physics-informed learning	Dynamics constrained by physical laws or symmetry	ERGTrtranslates physics-inspired principles into operational constraints for relational computation, without asserting a literal physical system.
ERGT	A raw-token Ψ field and executable induced geometry	Unifies structure induction, typed transport, geodesic composition, and fail-closed measurement in one raw-input-to-answer path.

Table 2 relates the needs of a long relational chain to the corresponding architectural objects. The physics-derived vocabulary is used in an operational sense. Ψ denotes persistent token information, a world is a directed relation lens, action is path cost, the cone masks invalid propagation, GeoTransport transfers typed state, a geodesic is the least-cost valid path, and boundary measurement is the final acceptance test.

These objects are constitutive design choices inspired by physics, not claims of literal physical law. Their role is testable: equations define them, invariants monitor their numerical behavior, and frozen interventions reveal whether the trained model needs them. Continuous neural states construct the field and local evidence; admissibility, typed-register, and terminal contracts then decide whether those local pieces form a valid answer. A high answer score cannot override an invalid intermediate program.

The target inference path is

$$\begin{aligned} \text{raw symbols} &\rightarrow \Psi \rightarrow \text{world metrics} \rightarrow \text{typed event support} \\ &\rightarrow \text{geodesic closure} \rightarrow \text{typed transport} \rightarrow \text{terminal measurement.} \end{aligned}$$

No standard QKV self-attention, Transformer encoder, discrete target compiler, generic graph executor, or learned answer MLP appears in this path.

3.2 Raw input and the persistent Ψ field

The construction begins with a state rich enough to support relations without smuggling in the relation graph. Semantic content alone is not sufficient: repeated or similar symbols must remain distinguishable, yet identity cannot be an external pointer to the answer. The initial field therefore joins semantic content and position with an identity fibre computed from the same raw token. Every later geometric quantity descends from this shared initial condition.

Table 2: Conceptual derivation of the architecture.

Problem need	Design principle	Architecture component
Preserve token identity and content	Persistent state from raw input	Ψ information field
Represent several relation views	Complementary directed metrics	Eight world lenses
Block impossible propagation	Explicit admissibility	Finite-speed cone
Keep operators bound to state	Type-preserving transfer	GeoTransport and payload registers
Compose long paths	Dynamic programming over action	Min-plus geodesic closure
Reject unsupported candidates	Hard terminal constraints	Boundary measurement and abstention

For token i , raw symbol fingerprints are mapped to a train-only semantic state x_i and a collision-audited identity fibre u_i . Both are derived inside ERGT from the public raw-token tensors. The initial condition is

$$\Psi_i^{(0)} = f_\psi(x_i + p_i + \gamma f_{\text{id}}(u_i)), \quad (1)$$

where p_i is the position state. The identity term changes both the initial field and the metrics derived from it, but supplies no pointer, graph edge, condition label, or answer channel. At the public boundary, the model sees only raw token identifiers and a padding mask. Equation 1 captures the field-construction claim: relational structure must emerge from pairwise observables of the same persistent state built from the serialization seen by the baseline.

3.3 Field observables and adhesion

The token field still has to determine which pairs can support a relation. Rather than compressing that decision into one unconstrained edge logit, ERGT exposes several bounded factors: coherence, reconstructibility, local compatibility, order, and agreement with the bulk–boundary view. Together they turn the node-like state Ψ into candidate edge support while leaving the source of weak support visible.

The hidden field induces the bounded observables

$$\rho_i = d^{-1} \|\Psi_i\|_2^2, \quad (2)$$

$$C_{ij} = \frac{1}{2} [1 + \langle f_C(\Psi_i), g_C(\Psi_j) \rangle], \quad (3)$$

$$R_{ij} = \sigma(2 \langle f_R(\Psi_i), g_R(\Psi_j) \rangle), \quad (4)$$

$$G_{ij} = \exp(-|\rho_i - \rho_j|), \quad (5)$$

$$B_{ij} = \exp[-(\delta_i^{\text{bulk}} + \delta_j^{\text{bulk}})/2]. \quad (6)$$

C measures coherence, R reconstructibility, G density-gradient compatibility, and B bulk–boundary agreement. Local relational entropy S_i gives the order term $O_i = 1 - S_i$, while stability is measured across field updates. The factors meet in an adhesion field formed by a geometric mean in log space:

$$\Phi_{ij} = \exp\left(\frac{1}{6} \sum_{k=1}^6 \log(z_{ij}^{(k)} + \epsilon)\right), \quad z^{(k)} \in \{C, R, \text{stability}, G, O, B\}. \quad (7)$$

Because the factors combine multiplicatively, one weak term cannot be hidden by several strong ones. Φ is an edge-formation gate chosen for this model, not a universal law of information.

Table 3: Operational world lenses. These are constitutive functions, not independently supplied graphs.

World	Dominant factors
W_0	locality, density-gradient fit, stability
W_1	coherence, identity overlap, local order
W_2	reconstruction and boundary reconstructibility
W_3	stability, identity overlap, locality
W_4	long bridge, coherence/identity support, reconstruction
W_5	density contrast and stability
W_6	novelty and long-bridge support
W_7	density source–sink flow and coherence

3.4 Constitutive world geometries

Adhesion establishes joint support, but not a single universal notion of proximity. Locality, reconstruction, long bridges, density flow, and boundary compatibility may favor different directed edges. Collapsing them immediately would erase those alternatives, so ERGT applies eight constitutive lenses to the shared field and keeps their potentials separate until a typed role needs them. Table 3 summarizes the resulting worlds.

For world w , edge state evolves as

$$E_{wij}^{(t+1)} = \beta E_{wij}^{(t)} + (1 - \beta) \Phi_{ij}^{(t)} \Omega_{wij}^{(t)}, \quad \ell_{wij} = -\log(E_{wij} + \epsilon). \quad (8)$$

Typed computational roles condition a router over these separate metrics. Let $\bar{\Psi}$ be the masked mean field, r a typed role, π_{rw} a learned role–world prior, and $\mathcal{W}_r$ the worlds authorized for that role:

$$\alpha_{rw} = \text{softmax}_{w \in \mathcal{W}_r} \left([W_{\text{route}} \bar{\Psi}]_{rw} + \pi_{rw} \right), \quad (9)$$

$$H_r = \sum_{w \in \mathcal{W}_r} \alpha_{rw} A_w(H_w), \quad (10)$$

where A_w adapts world w . A world is therefore not a separate predictor, but one directed metric through which the shared typed state can be viewed and transported. The worlds need not carry equal mass or have unique human-readable meanings. The testable claim is functional dependence on their composition: under frozen intervention, no single world completes the program.

3.5 Finite-speed admissibility and action

An edge potential says where a relation appears plausible; it does not yet make that edge executable. Keeping every nonzero affinity would leave weak, dense shortcuts through which the model could bypass a long relation chain. We therefore turn each learned world geometry into a sparse feasible domain and assign a non-negative cost to every legal local move. Admissibility determines *whether* state may propagate, while action distinguishes among the routes that remain.

Sparse support and a finite propagation condition define admissible edges:

$$\mathcal{A}_{wij} = \mathbf{1}[j \in \text{topk}(E_{wi.})] \mathbf{1}[\ell_{wij} \leq v_i \Delta \tau_{ij}]. \quad (11)$$

The path action is

$$S_{wij} = \ell_{wij} + \lambda_b(1 - B_{ij}) + \lambda_s(1 - \text{stability}_{ij}) + \lambda_c \text{coneViolation}_{ij}. \quad (12)$$

This action is a learned non-negative path cost. The term *finite-speed cone* means that the implemented update rejects edges whose effective travel cost exceeds the available discrete propagation interval; it is not a claim of relativistic spacetime.

3.6 GeoTransport and memory

A path cost is useful only if information can travel along the path and remain available to later steps. GeoTransport provides that state evolution. Its mask is inherited from admissibility and its weights from action, so an edge rejected by the geometric stage cannot reappear during transfer. Persistent momentum, edge state, and payload retain the history on which later geometry and terminal measurement depend.

Transport is induced by action over admissible neighbors:

$$T_{wij} = \text{softmax}_{j:\mathcal{A}_{wij}=1}(-S_{wij}/\tau), \quad (13)$$

$$M_{wi} = \sum_j T_{wij} V_w(\Psi_j), \quad (14)$$

$$\Pi_i^{(t+1)} = \mu \Pi_i^{(t)} + (1 - \mu) F(\Psi_i^{(t)}, M_i^{(t)}), \quad (15)$$

$$\Psi_i^{(t+1)} = \text{Norm}(\Psi_i^{(t)} + \Pi_i^{(t+1)}). \quad (16)$$

Memory is path-dependent state retained in E , Π , and transported payload. It can affect later geometry and terminal measurement but has no direct bypass to answer logits. In computational language, GeoTransport is therefore a masked, action-weighted transfer operator that keeps a typed state attached to the path on which it arrived; it is not attention-weighted value mixing.

3.7 Typed event support

Geometry determines where state may travel and what the journey costs, but it does not by itself identify the relation operator carried by an event. That semantic content must remain sufficiently discrete to prevent a short route from substituting one operator for another. Typed event support therefore binds source, relation, and target proposals to the path, while the geometry retains control over admissibility and route selection. This provides the interface between learned interpretation of the raw symbols and native geometric execution, without introducing a compiled target graph.

Predicted event source, relation, and target distributions form typed charge

$$Q_{ijr} = \sum_e p(i | e) p(r | \Psi_e^{(0)}) p(j | e). \quad (17)$$

The charge supplies typed local proposals to the geometric program. Geometry determines whether and at what cost a proposal can participate in a complete path; it cannot transmute the discrete operator type carried by Q . This separation prevents a low scalar path cost from silently changing the relation being transported.

3.8 Product-manifold geodesic closure

The preceding stages define legal one-step transitions. To reason over a long horizon, the model must compose them without handing the result to a fixed-depth answer classifier. ERGT does so by running dynamic programming over the discrete product of token location and world lens. One coordinate records where the state is; the other records the directed metric under which its current transition is evaluated. Min-plus closure can then extend the same local action contract to paths that exceed a single field update.

The solver state is the discrete product (i, w) of token and world. For candidate c , the min-plus

field is initialized at its registered source:

$$D_c^{(0)}(i, w) = \begin{cases} 0, & (i, w) \text{ is an allowed source,} \\ +\infty, & \text{otherwise,} \end{cases} \quad (18)$$

$$D_c^{(t+1)}(j, w') = \min \left\{ D_c^{(t)}(j, w'), D_c^{(t)}(i, w) + A[(i, w) \rightarrow (j, w')] \right\}. \quad (19)$$

Support is propagated by a max–min widest-path channel so repeated probability multiplication alone cannot erase a valid long path. Closure depth is at least the registered answer horizon and is capped at 36 steps in the paper profile. A label-free dyadic valid-domain mesh allows action to propagate over long raw sequences without supplying the answer topology. Here *geodesic* has a precise discrete meaning: the path of minimum total implemented action among paths that satisfy the admissibility mask.

3.9 Typed payload and terminal decision

A least-action route establishes reachability, but reachability alone is not an answer. The arriving state must also encode the operator sequence required by the candidate, preserve enough transported support, and satisfy the registered terminal boundary. The final stage couples path optimization to typed register execution and asks, through hard measurements, whether the assembled evidence is complete. This is where fail-closed behavior becomes part of the computation rather than an output convention.

Along an admissible event, the register and payload evolve as

$$r' = \text{applyOperator}(r, r_e), \quad (20)$$

$$q^{(t+1)}(j, r') = \max_e q^{(t)}(i_e, r) \tau_e m_e, \quad (21)$$

$$\Delta^{(t+1)}(j, r') = \min_e \left[\Delta^{(t)}(i_e, r) + \delta_e \right]. \quad (22)$$

Two normalization contracts are checked before checkpoint eligibility:

$$\varepsilon_{\text{typed}} = \max_e \left| \sum_r p_e(r) - p_e(\text{present}) \right|, \quad (23)$$

$$\varepsilon_{\text{transport}} = \max_{w, i \text{ valid}} \left| \sum_j T_{wij} - 1 \right|. \quad (24)$$

The first keeps relation-type mass tied to event-presence mass; the second keeps each valid transport row normalized. For candidate c , let A_c be its closed geodesic action, m_c its terminal payload mass, and δ_c its boundary deficit. Its hard validity indicator is

$$V_c = \mathbf{1}[A_c < \infty] \mathbf{1}[m_c \geq m_{\min}] \mathbf{1}[\delta_c \leq \delta_{\max}], \quad (25)$$

where A_c is read from the register state required by that candidate, so a wrong typed register is unsupported. Cone validity is already enforced in the edges from which finite A_c can be assembled. The decision is

$$\hat{c} = \arg \min_{c: V_c=1} A_c \quad \text{only if } A_{(2)} - A_{(1)} \geq \eta; \quad \text{otherwise abstain.} \quad (26)$$

Thus boundary measurement means checking the final register, mass, conservation, and deficit before accepting a candidate. Two unresolved candidates, no valid candidate, or insufficient action margin produces abstention. This is a native geometric boundary solver, not a call to a generic graph-traversal routine.

Table 4: Inferential structure of the registered study.

Question	Evidence panel	Permitted conclusion
Was the inference path learned?	Train ID, raw-disjoint validation, and native stability contracts	Model-specific convergence and, for ERGT, executable-path readiness
Does computation extend in horizon?	Frozen matched cohorts beyond the training range	Bounded long-horizon accuracy and paired model difference
Does the model fail closed?	Controlled inputs with no valid candidate	Rejection within the registered unsupported protocol
Is geometry functionally necessary?	Same-checkpoint removals, corruption, and single-world controls	Causal attribution to the intervened mechanism on eligible solved pairs
How does geometry change with horizon?	Detached curvature, spectrum, and world-diversity observers	Descriptive structure only, not causal performance attribution

3.10 Training objective and zero-teacher verification

Training must reconcile differentiable field construction with hard execution contracts. The upstream field needs enough supervision to populate the native path, yet the reported inference cannot depend on a teacher or a soft answer bypass. We therefore group intermediate losses by architectural role and keep checkpoint selection separate from the subsequent frozen execution test.

The grouped training objective is

$$\mathcal{L} = \mathcal{L}_{\text{answer}} + 2a(t)\mathcal{L}_{\text{structured}} + 2a(t)\mathcal{L}_{\text{physical}} + 2a(t)\mathcal{L}_{\text{causal}} + 2a(t)\mathcal{L}_{\text{margin}} + 0.5\mathcal{L}_{\text{geometry}}. \quad (27)$$

$a(t)$ is a teacher schedule. Structured terms supervise entity, query role, event type, endpoints, operator, boundary, and edge assembly. Physical terms supervise action, cone, transport, boundary deficit, and terminal mass. Geometry terms include bulk-boundary reconstruction, conservation, world usage, and diversity. A checkpoint becomes eligible only after independent field-readiness windows. Teacher weight is then set to zero, parameters are frozen, and two further windows must pass with zero optimizer steps. Thus the reported answer is teacher-free at inference, although training is supervised.

4 Experimental Design

The experiment mirrors the computational chain defined in Section 3. It first asks whether raw symbols can produce a stable native geometry, then whether the frozen computation survives longer relation chains, and finally whether its answers depend on the geometric operations claimed by the method. Unsupported inputs add a separate test of whether the terminal rule can withhold an answer. These questions cannot be answered by a single accuracy score, so each is assigned its own evidence panel.

The panels are also kept separate during model selection. Development data are used only to lock the strengthened direct-Transformer recipe; native readiness data select the first stable ERGTcheckpoint; and the final horizon cohorts remain unopened until both procedures are complete. Interventions are applied only to that frozen native checkpoint. Table 4 states the resulting inferential boundary.

4.1 Benchmark and counterfactual cohorts

The benchmark isolates relational composition from open-domain language knowledge. Each example serializes nodes, relation events, a query, two candidates, and distractors as raw symbols; this serialization is the complete input to both models. A latent graph exists only inside the generator so that

examples can be constructed and matched. Adjacency, edge indices, entity pointers, and candidate paths are never exposed.

The basic unit is an answer-reversing counterfactual pair. Its members share topology, edge count, hop count, operator types, and lexical statistics, but one registered condition changes the valid candidate. Scenarios separately vary geodesic action, finite-speed admissibility, boundary deficit, payload transport, terminal mass, source memory, and their combined long-horizon use. The pair design therefore rewards tracking the condition that governs a path, not memorizing its surface form or candidate position.

Training contains 28 pair groups at each hop from 1 through 8: 224 supported groups in total, plus 28 controlled unsupported groups. Independent native readiness data cover hops 2–8. Frozen evaluation uses 4, 8, 12, 16, 24, and 32 hops; context panels use 128, 256, 512, and 768 tokens; and unsupported cases are tested at 4, 16, and 32 hops. The horizon panels vary relational depth, whereas the context panels add serialized material around a fixed decision. This controlled synthetic setting is intended to isolate the mechanism and does not measure open-domain language ability.

4.2 Compared inference paths

Fairness is defined at the external task boundary rather than by forcing the architectures to share an internal solver. Both receive the same serialization and padding mask, train on the registered 1–8-hop family under locked schedules, and predict candidate A, candidate B, or abstention. The baseline is strengthened during training, but no ERGT-derived structure enters its inference path. The comparison is therefore between two complete ways of deriving an answer from raw symbols, not between ERGT and a Transformer equipped with the ERGTSolver.

The direct arm is a standard PyTorch Transformer encoder with token embeddings, sinusoidal positions, QKV multi-head attention, GELU feed-forward blocks, layer normalization, and a three-class answer head. Its locked configuration uses width 64, four heads, four layers, feed-forward width 192, context 768, and zero dropout. It has 205,634 inference-active parameters, compared with 196,816 for ERGT, a 4.48% difference. Matched auxiliary targets strengthen training but are not inference inputs, and the model receives no graph, geometric state, condition side input, compiler, program, executor, or identity side channel.

Native ERGT uses width 56, Ψ rank 20, eight world lenses, three field updates, sparse top-20 support, and at most 36 geodesic-closure steps. From the same serialization and mask, it must construct identity, relation support, world geometry, typed transport, and the terminal decision through the path in Section 3; no learned answer MLP bypasses that route. Parameter proximity limits a simple capacity-scale explanation, but it does not imply equal computation. Optimizer steps, time, and peak CUDA allocation are therefore measured rather than matched.

4.3 Locked training and checkpoint selection

The Transformer recipe was qualified once on development seeds before any 12–32-hop final cohort was opened. It begins with one-hop warm-up at batch size 8 and learning rate 7×10^{-4} ; after two eligible raw-disjoint ID windows, it may enter fixed stages with exact 1–2, 1–4, and 1–8-hop coverage. Architecture, optimizer, transition rule, and step budget are then locked. A missed transition is recorded rather than repaired using final data.

The confirmatory `four_seed_confirmation` profile trains both models from scratch. Optimization seeds 17117, 18313, 19501, and 20707 are paired with independent data seeds 21101, 22307, 23509, and 24733. All raw sets are collision-audited and disjoint across training, tuning, validation, and final evaluation. Horizon, context, unsupported, intervention, and observer panels remain outside checkpoint selection.

During confirmation, the Transformer follows its locked warm-up and curriculum within 1400 steps. ERGT uses the registered 1–8-hop pool, batch size 8, AdamW at 2.5×10^{-3} , and at most 1800 steps. Its teacher is used only during training. The first checkpoint to complete two 2–8-hop field-readiness windows is frozen immediately, then evaluated in two unchanged zero-teacher windows without another optimizer update. Selection therefore depends on a stable short-to-intermediate native program, never on the final horizon score.

4.4 Registered outcomes and interventions

Row accuracy records individual answers; counterfactual-pair exactness requires both members of an answer-reversing pair to be correct. Model-specific convergence requires at least 0.90 supported accuracy on both train ID and untouched raw-disjoint validation. A secondary matched-convergence analysis is restricted to seed pairs for which both models satisfy that rule. Failure in one arm limits only joint-convergence claims and does not erase the other model’s independent result.

The preregistered primary comparison is the 32-hop panel. Its bounded claim requires native accuracy of at least 0.90, a margin of at least 0.35 over the direct Transformer, a hierarchical bootstrap interval excluding zero, and exact paired McNemar $p < 0.05$. Counterfactual pairs are the paired clusters and are nested within data seed. Bootstrap intervals follow Efron and Tibshirani [20]; discordant outcomes use McNemar’s exact test [21]. Intermediate horizons, context lengths, pair exactness, and unsupported rejection remain supporting panels and cannot select a checkpoint.

Native checkpoint eligibility is stricter than answer accuracy. Measurement accuracy must reach 0.99; action MAE must be at most 0.12; event recall, precision, path coverage, and chain exactness must reach 0.99, 0.98, 0.99, and 0.99; closure must not degrade; hard overflow must be zero; conservation residual must be at most 10^{-4} ; every group must have exposure 8; and two consecutive readiness windows must pass. Batch, padding, identity, and raw-input invariance are recorded with these contracts. All registered outcomes remain in the readout even when a claim-specific rule is not met.

Mechanism attribution then reuses the selected checkpoint without retraining. It changes one inference operation at a time and evaluates only counterfactual pairs solved by the intact model. Targeted removals disable action, cone, typed transport, boundary deficit, terminal mass, or source-conditioned memory; global controls randomize or shuffle geometry, bypass the conditioned world gate, or retain one world. The minimum targeted drop is 0.20. Formal evaluation requires intact accuracy of at least 0.80 and at least two eligible pairs; conditional effects are still reported below that panel-wide floor, but the corresponding claim is narrowed.

Curvature, graph gradient, Laplacian spectrum, spectral entropy, effective rank, spectral gap, and world diversity are handled differently. They observe the frozen geometry outside the answer path and cannot change a prediction. Their role is descriptive and prospective, not causal attribution for current performance.

5 Results

The evidence follows the life of a single learned computation. We first verify that the native path converges and can be frozen without losing its contracts. That fixed path is then carried to greater relational depth and longer contexts before the same checkpoints face unsupported inputs, mechanism interventions, and detached geometric observation. Fitting, extrapolation, and causal attribution thus remain separate empirical questions, but their answers refer to the same computation.

Table 5: Accuracy by independent optimization/data-seed pair. Values are from fresh confirmatory runs.

Seed pair	ERGTtrain	ERGTval.	TRANSFORMERtrain	TRANSFORMERval.	Freeze step
17117/21101	1.000	1.000	1.000	0.510	700
18313/22307	1.000	1.000	1.000	0.459	700
19501/23509	1.000	1.000	1.000	0.837	1000
20707/24733	1.000	1.000	1.000	0.449	1000

Table 6: Mean accuracy across four independent optimization/data-seed pairs. Training groups cover 1–8 hops.

Hops	Native ERGT	Direct Transformer	Difference
4	1.0000	0.4821	0.5179
8	1.0000	0.5045	0.4955
12	0.9821	0.5000	0.4821
16	0.9866	0.5000	0.4866
24	0.9196	0.5000	0.4196
32	0.9554	0.5000	0.4554

5.1 Convergence and checkpoint integrity

The long-horizon study begins with the computation that will be extrapolated. As Table 5 shows, native ERGT reached 1.000 accuracy on both training and raw-disjoint validation in all four optimization/data-seed pairs. The complete route—from Ψ formation through world geometry and GeoTransport to terminal measurement—therefore converged across fresh data and initializations. The strengthened direct Transformer also reached 1.000 on every training split, but its held-out accuracy ranged from 0.449 to 0.837. This gap describes the registered baseline; it does not alter the independent convergence result for ERGT.

Accuracy was not sufficient to select a checkpoint. The first state to pass two complete readiness windows was frozen only after every registered 2–8-hop group had sufficient exposure. The four freeze steps were 700, 700, 1000, and 1000; minimum group exposures were 11, 11, 15, and 15 against the required floor of 8. No group was unseen, and no optimizer update followed freeze.

The internal audits reach the same conclusion at finer resolution. All 56 instances of the 14 native stability invariants passed. Event-chain exactness, geodesic path coverage, closure non-degradation, event precision, and event recall were each 1.000; overflow was zero; and the largest typed-payload conservation residual was 2.38×10^{-7} . Batch-size and padding invariance, identity construction from raw input, and raw-input parity also passed. The selected states were therefore stable executions of the intended native path rather than products of checkpoint drift, an external identity channel, or serialization artifacts.

5.2 Frozen generalization across horizon and context

Those states were then held fixed throughout the horizon study. Because training groups covered 1–8 hops and no final panel could influence selection, Table 6 measures extrapolation rather than adaptation. Mean ERGT accuracy remained above 0.90 at every registered horizon and reached 0.9554 at 32 hops, whereas the direct-Transformer means remained near 0.50.

The aggregate result includes visible seed variation. At 32 hops, ERGT accuracy was 0.982, 1.000, 1.000, and 0.839 across the four runs; exactness on the corresponding answer-reversing counterfactual

Table 7: Mean correct rejection rate on controlled unsupported candidates.

Hops	Native ERGT	Direct Transformer
4	1.0000	0.0000
16	1.0000	0.0000
32	1.0000	0.0000

pairs was 0.964, 1.000, 1.000, and 0.750. The pooled accuracies are

$$\text{Acc}_{\text{ERGT}} = 0.955357, \quad \text{Acc}_{\text{TRANSFORMER}} = 0.500000.$$

Their paired difference, 0.455357, exceeds the registered 0.35 margin, while the native mean exceeds its 0.90 floor. A hierarchical bootstrap over data seed and counterfactual pair gives the 95% interval $[0.375, 0.500]$, and the exact paired test gives

$$p_{\text{McNemar}} = 9.86 \times 10^{-32}.$$

Of the 224 examples in 112 pairs, 104 were correct only for ERGT and none only for the Transformer, matching the direction of the bootstrap interval and exact test. Within this benchmark, the frozen geometric computation therefore retains substantially more accuracy as relational depth extends to 32 hops.

The context panels distinguish that effect from tolerance of extra tokens. At hop 4, mean ERGT accuracy was 1.000, 1.000, 1.000, and 0.963 at 128, 256, 512, and 768 tokens, compared with 0.500, 0.506, 0.519, and 0.500 for the Transformer. At hop 24 and contexts of 512 and 768 tokens, the corresponding values were 0.931 and 0.988 for ERGT versus 0.519 and 0.531. The horizon effect thus persists inside the registered longer serializations, without being presented as a general long-context language-modeling result.

5.3 Fail-closed behavior and geometric attribution

The next question is not how often the model answers, but whether it answers only when its own terminal conditions are satisfied. On controlled cases in which neither candidate has admissible transported support, ERGT rejected every example at 4, 16, and 32 hops in every seed (Table 7); the direct Transformer rejected none.

This is the benchmark’s fail-closed result: absence of valid geometric support produces abstention rather than a forced choice. It is not an open-domain hallucination measure.

The same frozen checkpoints were then used to test whether successful answers actually depend on the proposed geometry. Table 8 reports scenario-matched interventions on pairs solved by the intact model, so each drop is the loss of an established capability. Action ranks admissible paths, the cone excludes invalid propagation, transport carries typed state, terminal mass verifies its arrival, memory preserves source information, and boundary deficit checks terminal compatibility. Removing action, cone, transport, or terminal mass reduced eligible accuracy from 1.000 to 0.000 in all four seeds; removing memory left 0.031.

Boundary removal likewise changed eligible solved-pair accuracy from 1.000 to 0.000 in every seed. Formal panel-wide qualification was 2/4 because two intact boundary panels scored 0.750, below the conservative 0.80 evaluability floor. The conditional effect is therefore present in all four runs, while the formal panel claim remains limited to the two preregistered eligible seeds.

The global controls lead to the same interpretation. Random geometry reduced eligible accuracy by 0.646, cross-example shuffling by 0.597, and bypassing the conditioned world gate by 0.382. Restricting

Table 8: Frozen mechanism interventions. Full and intervention accuracies are computed on eligible solved pairs; “formal seeds” also satisfy the panel-wide 0.80 intact-model floor.

Removed mechanism	Full	Intervention	Formal seeds
Geodesic action	1.000	0.000	4/4
Finite-speed cone	1.000	0.000	4/4
Typed payload transport	1.000	0.000	4/4
Terminal mass	1.000	0.000	4/4
Source-conditioned memory	1.000	0.031	4/4
Boundary deficit	1.000	0.000	2/4

Table 9: Detached geometric observers, averaged over four seeds.

Observer	4 hops	32 hops
Mean curvature proxy	−0.1011	−0.0154
Negative-curvature fraction	0.8959	0.9479
Mean absolute curvature gradient	0.002424	0.000354
Spectral effective rank	4.859	43.867
Spectral entropy	0.711	0.868
Spectral gap	0.1512	0.0115
World diversity	0.1447	0.8052

inference to any one of worlds 0–7 produced zero accuracy. The result thus depends on the learned arrangement of costs and constraints and on composition across worlds, without requiring equal world use or a unique human-readable meaning for each world.

5.4 Detached geometric observers

Curvature and spectrum were observed alongside the answer path, never used as answer features. The detached CPU/float64 pipeline processed 26,176 induced graphs without unavailable values, sanitization, numerical failure, or a CUDA eigensolver call. In Table 9, curvature describes local graph expansion and contraction; spectral rank, entropy, and gap summarize global mode use and connectivity; and world diversity records how broadly the geometric lenses participate.

Between 4 and 32 hops, effective spectral rank rose from 4.859 to 43.867, entropy from 0.711 to 0.868, and world diversity from 0.1447 to 0.8052. Spectral gap fell from 0.1512 to 0.0115 and mean absolute curvature gradient from 0.002424 to 0.000354, while the Forman-style curvature proxy remained predominantly negative. The geometry therefore becomes more distributed across modes and worlds as chains lengthen. Because the observers were detached, this is a structural description, not evidence that curvature or spectrum caused the accuracy gain.

5.5 Computational cost

The reported behavior carries a higher computational cost. Mean training time on the Colab runtime was 1437.5 seconds for ERGTand 56.25 seconds for the direct Transformer, a per-seed ratio of 21.3–29.8. Mean peak CUDA allocation was approximately 3.432 GiB versus 1.455 GiB, a ratio of 2.36. These values are reported for transparency and do not support an efficiency claim.

Read as one sequence, the four-seed evidence connects convergence of the native raw-token-to- Ψ path to frozen 32-hop generalization, fail-closed behavior, and mechanism-specific intervention effects. The numerical advantage is thus paired with evidence that the learned geometry, transport, memory, and terminal constraints participate in the answer, at the greater resource cost reported above.

6 Discussion

The results support a direct conclusion: ERGT learned the complete route from raw tokens to a constrained decision. The Ψ field was not an intermediate representation handed to a conventional readout. It supplied the identity and relation evidence from which the model formed directed world geometries, transported typed state, and licensed a terminal answer. This route converged in every fresh run and remained intact after the checkpoint was frozen. The experiment therefore establishes more than the feasibility of writing down a geometric model; it shows that the proposed geometry can be learned end to end and used as the model’s native inference path.

The horizon panels then ask whether that learned computation is reusable. They were never used to choose a checkpoint, so the performance reported at long relation depths reflects a frozen rule learned from shorter compositions rather than adaptation to the test horizon. The context controls help locate the effect: increasing the number of surrounding tokens is not equivalent to increasing the number of relations that must be composed. Across the registered panels, the geometric path remained strongest when the task required deeper composition, while the unsupported cases showed that it could also withhold an answer when the terminal conditions were not met.

The direct-Transformer comparison gives this result a clear, limited meaning. Both models start from the same raw serialization, receive matched training labels, and have closely matched active parameter counts. The Transformer was trained with the strengthened, locked recipe selected during development, but it received no ERGT geometry or solver at inference. It fitted the fresh training sets yet did not carry the same rule across raw-disjoint validation and longer horizons. Under this protocol, explicit construction and execution of a relation geometry generalized better than direct content mixing. This is an architectural result for the registered benchmark, not a claim that every Transformer must fail or that the comparison represents universal post-convergence superiority.

The interventions explain why the accuracy difference is attributable to the proposed computation. Removing path action, propagation admissibility, typed transport, terminal mass, or source-conditioned memory broke distinctions that the intact model solved. Random and shuffled geometries also caused substantial losses, and no single world reproduced the complete computation. These results rule out a simple account in which the geometric quantities are merely labels attached to an unrelated predictor. They support the mechanism as a composed system: the evidence does not require every component to be sufficient on its own, but it does show that the registered path depends on its geometry, transport, memory, and terminal constraints.

This dependence also offers a natural explanation for the long-horizon behavior. Ψ preserves identity and local relation evidence; the worlds retain distinct directional views; the cone removes inadmissible propagation; action ranks the remaining paths; GeoTransport carries the typed payload and memory; and terminal measurement decides whether the transported state is a valid answer. Extending a chain repeats these local contracts without changing their meaning. The observed horizon generalization is therefore consistent with reuse of a stable geometric law rather than reconstruction of a new global correlation at each length. The frozen interventions turn that interpretation into a testable account rather than a metaphor applied after prediction.

The broader contribution is consequently not the absence of attention by itself. It is the construction of task-relevant structure from raw input and the immediate use of that structure as an executable, fail-closed geometry. The model is not given a graph, and the induced graph is not retained only as an explanation: its support, direction, cost, transported state, and boundary conditions determine the answer. This places ERGT between sequence modeling, latent structure learning, path reasoning, and physics-informed design, while the combination tested here remains distinct from simply replacing the sequence mixer or running a learned algorithm on a supplied graph.

Finally, the curvature and spectral measurements describe how the induced geometry changes with depth, but they do not participate in the present decision path. Their trends suggest concrete future

tests: curvature may help regularize action or detect transport bottlenecks, while spectral rank, gap, and entropy may help monitor world allocation, collapse, memory organization, or structural shift. Those possibilities remain prospective. The result established here is narrower and firmer: within the registered relational family, a raw-token information field can support a convergent native geometry, and that geometry can sustain auditable long-horizon reasoning.

7 Scope and Limitations

This study addresses a bounded architectural hypothesis: geometry induced from raw tokens can execute long-horizon relational reasoning. Internal validity within the registered synthetic family is supported by four independent seed pairs, fresh data, frozen checkpoints, paired counterfactuals, invariant audits, and same-checkpoint interventions. Whether the result extends to natural language, broader tasks, or other model scales remains open.

The parameter-matched direct Transformer was strengthened and locked before the confirmatory runs, but did not reproduce its development held-out criterion on the fresh data. The comparison therefore establishes an architectural advantage for this benchmark, not universal or post-matched-convergence superiority. Training uses structured supervision, whereas native inference is teacher-, compiler-, and executor-free. Boundary dependence is exact on eligible solved pairs and meets the formal panel-wide criterion in two of four seeds. Curvature and spectrum remain descriptive observers, computational efficiency is not claimed, and unsupported rejection refers only to the registered protocol.

8 Code and Data Availability

The complete reproduction release is available in the public [ERGT-paper](#) repository. To run it, first choose **Code > Download ZIP** on that page and extract the downloaded `ERGT-paper-main.zip`. Then open the [executable Colab notebook](#), select a GPU runtime, and choose **Run all**. When the notebook opens the upload dialog, select the inner archive `ERGT-paper-main/ERGT_FourSeed_Reproduction.zip` from the extracted folder. The outer `ERGT-paper-main.zip` is not the runtime package and must not be uploaded to the notebook.

9 Conclusion

ERGT demonstrates a distinct route from sequence input to relational reasoning. Instead of using self-attention or a Transformer backbone to mix the sequence and read out an answer, it organizes raw-token information into a persistent Ψ field. Identity and relation evidence in that field induce multiple directed geometries; GeoTransport carries typed state through their admissible geodesics; and boundary measurement accepts an answer only when the transported state satisfies the terminal constraints. The induced geometry is therefore neither a supplied graph nor a post-hoc explanation. It is the model’s native inference machinery.

This complete field-to-answer path converged in four independent runs, passed all 56 stability invariants, and retained mean accuracy 0.955 at 32 hops, exceeding the strengthened direct Transformer by 0.455 with strong paired evidence. Frozen interventions further tied the solved answers to geometric action, admissibility, transport, memory, terminal mass, and multiworld composition. Within the registered relational family, these results establish that a trainable, attention-free information geometry can form from raw input, converge end to end, and reuse its local transport laws for substantially longer chains. This is the central contribution and the basis for studying executable information fields as an alternative architectural direction for long-horizon reasoning.

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A Operational Physics Glossary

Table 10: Reader-facing interpretation of physics-derived terms.

Term	Implemented meaning	Scope
Ψ field	Token-indexed persistent neural state initialized from raw semantics, position, and internal identity fibre.	Not a physical wavefunction or literal field.
Density ρ	Mean squared field amplitude per token.	Neural diagnostic, not mass or energy density.
Adhesion Φ	Multiplicative gate over coherence, reconstruction, stability, gradient, order, and boundary factors.	Constitutive modeling choice, not a universal information law.
World lens	One of eight functions mapping the shared field to a directed edge potential.	Functional multiworld dependence is tested; unique human semantic roles are not claimed.
Metric/action	Non-negative learned path cost derived from world edge strength and penalties.	Discrete path cost, not a proved smooth Riemannian metric.
Finite-speed cone	Hard admissibility condition based on path cost and a discrete propagation interval.	Causal computational constraint, not physical relativity.
GeoTransport	Geometry-conditioned movement of field and typed payload.	Not standard QKV value mixing.
Product manifold	Discrete Cartesian state (token, world).	A product graph state space, not a proved differentiable manifold.
Geodesic	Minimum-action path computed by min-plus closure.	Shortest path in the induced discrete action, not a continuum geodesic theorem.
Boundary deficit	Candidate-specific mismatch with registered terminal conditions.	Hard decision observable, not a physical boundary measurement.
Boundary measurement	Final conjunction of typed register, terminal mass, deficit, reachability, and action-margin tests.	Computational answer acceptance or abstention rule.
Terminal mass	Transported support arriving at a valid terminal state.	Payload support, not physical mass.
Memory geometry	Persistent path-dependent edge and payload state.	Bounded task memory, not general episodic or language memory.
Curvature proxy	Detached Forman-style edge diagnostic.	Observer only; no causal benefit claimed.

Term	Implemented meaning	Scope
Spectrum	Eigenvalue diagnostics of a geometry-derived graph Laplacian.	Observer only; no performance role claimed.